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Training Report
AI - Optic Disc & Cup Segmentation
v1.0

U-Net++ Training Results (Baseline, No Augmentation)

Project Subject: Glaucoma detection using intelligent analysis of fundus images

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1 Introduction

This report documents a baseline training run of the U-Net++ optic disc and cup segmentation model performed **without data augmentation**. It serves as a reference point against which the effect of augmentation and other improvements can be measured.

2 Experimental Setup

- **Architecture:** U-Net++ with an `efficientnet-b4` encoder (20.8M parameters).
- **Device:** CUDA (GPU).
- **Dataset:** 1020 fundus images — 816 training / 204 validation.
- **Epochs:** 50.
- **Primary metric:** Dice score (validation).
- **Augmentation:** none (baseline run).

Note. The run logged a warning that the argument `std_dev` is not valid for the `GaussNoise` transform; since this is a no-augmentation baseline, it did not affect the reported results.

3 Results

The best validation Dice score was **0.8940**, reached at epoch 32. The table below lists representative epochs (each row where a new best model was saved, plus the final epoch).

Epoch	Train Loss	Train Dice	Val Loss	Val Dice
1	0.4282	0.7546	0.3547	0.8274
2	0.2670	0.8548	0.2506	0.8600
5	0.1624	0.8860	0.1874	0.8761
10	0.1283	0.9065	0.1685	0.8848
13	0.1078	0.9172	0.1714	0.8870
19	0.0872	0.9297	0.1661	0.8876
25	0.0719	0.9397	0.1659	0.8897
32	0.0593	0.9485	0.1561	0.8940
50	0.0454	0.9581	0.1583	0.8930

Table 1: Selected epochs from the baseline U-Net++ run. Best validation Dice in bold.

4 Discussion

Training Dice keeps rising to ≈ 0.958 by epoch 50, while validation Dice plateaus around 0.89 after epoch 25 and the validation loss stops improving. This growing gap between training and validation is a clear sign of **overfitting** — expected for a relatively small dataset (1020 images) trained without augmentation. The best checkpoint (epoch 32, Val Dice 0.8940) was saved as `unetpp_best.pth`.

5 Summary

The no-augmentation baseline reaches a validation Dice of **0.8940** but overfits in the later epochs. This motivates the improvements proposed in the accompanying document (*Proposed Improvements to the Glaucoma Segmentation Model*), in particular data augmentation, regularisation, and a corrected validation transform, which are expected to close the train/validation gap and push the Dice score toward the $\geq 95\%$ target.